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MSc in Computer Science

Data, Information, and Knowledge Management

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Group project proposal

From Hormuz to the Boarding Gate: Visualizing Oil Shocks and Airline Ticket Prices

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Abstract

This project visualizes how disruptions around the Strait of Hormuz can propagate from global oil markets to airline ticket prices. Using interactive dashboards, we trace links between conflict events, Brent crude, jet fuel costs, fuel surcharges, fare changes, and route disruptions, helping non-experts understand how geopolitical shocks affect passengers directly.

Background and Motivation

The Strait of Hormuz, between Oman and Iran, is the world's most critical oil chokepoint. According to the U.S. Energy Information Administration, more than one-quarter of all global seaborne oil trade and roughly one-fifth of global LNG trade pass through it, with very few alternative shipping routes. Even short disruptions tend to translate into oil price spikes within days.

This is not a hypothetical scenario. As of April 2026, the Strait is operating under heavy restrictions amid escalating regional tensions, and Brent crude has surged toward \$105 per barrel. Higher crude prices push up jet fuel, which is typically the largest single operating cost for airlines, and airlines respond through fuel surcharges and higher base fares. At the same time, some routes are rerouted around closed airspace, adding distance and fuel burn.

Our motivation is to make this chain reaction visible. The link between a geopolitical event happening 4,000 km away and the price stamped on a boarding pass is intuitive to almost no one, yet it shapes the cost of flying for everyone. We believe a focused visualization can communicate this far better than text or statistics alone.

Project Aims and Research Questions

The main aim is to design an interactive web-based dashboard that lets a broad audience trace, step by step, how a Strait of Hormuz disruption propagates into the cost of flying.

Key questions:

1. How do Brent crude and jet fuel prices respond during identified conflict phases compared to baseline periods?
2. How quickly, and how strongly, do airlines adjust ticket prices and fuel surcharges after fuel cost shocks?
3. Which airlines, regions, or route classes appear most exposed to fuel price shocks?
4. What share of the final ticket reaches the passenger as base fare, fuel surcharge, and taxes?
5. How do route-level disruptions, such as rerouting, extra distance, or cancellations, compound the cost effect?

Target Audience

Our target audience is broad and non-specialist: travelers who want to understand why tickets become more expensive, students of data visualization, energy, or geopolitics, and journalists or general readers looking for a visual explanation of how global events affect everyday consumer prices. The dashboard must be understandable without prior knowledge of aviation economics or oil markets.

Datasets Considered

Primary dataset (core mechanics of the dashboard):

- **Airline Ticket Prices vs Oil and Fuel Costs** (Kaggle, by Z. K. S. Khurram). A research-grade synthetic dataset calibrated to real-world benchmarks, covering 2019 to 2026. It includes six tables: oil_jet_fuel_prices, conflict_oil_events, airline_ticket_prices, fuel_surcharges, route_cost_impact, and airline_financial_impact. Together they cover roughly 14k airline-month observations, 88 monthly oil/fuel snapshots, 37 conflict events, and 3,100+ route-level records. We treat this dataset transparently as scenario-based, not as factual history.

Download: <https://www.kaggle.com/datasets/zkskhurram/airline-ticket-prices-vs-oil-and-fuel-costs>

Supplementary datasets (real-world context and validation):

- **EIA, Strait of Hormuz chokepoint analysis.** Used for the introductory map and context panel.
<https://www.eia.gov/todayinenergy/detail.php?id=65504>
- **IEA, Strait of Hormuz oil security page.** Complementary chokepoint context.
<https://www.iea.org/about/oil-security-and-emergency-response/strait-of-hormuz>
- **FRED, Brent Crude Oil Price (DCOILBRENTU).** Real Brent series from 1987 onward, used as a real-world overlay against the synthetic series. Direct CSV download available.
<https://fred.stlouisfed.org/series/DCOILBRENTU>
- **FRED / EIA, Kerosene-Type Jet Fuel Prices, U.S. Gulf Coast (DJFUELUSGULF).** Real jet fuel benchmark from 1990 onward.
<https://fred.stlouisfed.org/series/DJFUELUSGULF>
- **OPEC, Monthly Oil Market Report (MOMR).** Used for production data and market commentary supporting the oil-shock narrative.
<https://www.opec.org/monthly-oil-market-report.html>
- **OurAirports, public airport database.** Direct CSV download of airport coordinates and IATA codes, used to build the route and disruption map.
<https://ourairports.com/data/>

- **ACLED, Armed Conflict Location & Event Data.** Geocoded conflict events for cross-referencing the Middle East timeline with our synthetic conflict-event table. Free download after account registration.
<https://acleddata.com/conflict-data/download-data>
- **U.S. BTS, Airline Origin and Destination Survey (DB1B).** Quarterly U.S. airfare microdata, used as a real-world airfare comparison.
<https://www.bts.gov/topics/airlines-and-airports/origin-and-destination-survey-data>
- **Eurocontrol, Aviation Intelligence Portal.** European flight movements and delay data, used to support the route-disruption view with real European traffic context.
<https://www.eurocontrol.int/our-data>
- **World Bank, Air Transport Passengers Carried (IS.AIR.PSGR).** Annual passenger volumes per country, used to weight regional impact and provide industry context.
<https://data.worldbank.org/indicator/IS.AIR.PSGR>

We will use the Kaggle dataset as the integrated backbone of the dashboard, and the official sources above for context, real-world overlays, and credibility, without attempting a full physical merge across all of them.

Data processing will be carried out in Python using pandas. Key steps include converting month strings to proper date objects, computing year-over-year and shock-window differences for prices and fares, joining route_cost_impact records to OurAirports coordinates by IATA code for geographic plotting, and tagging each month as a "disruption" or "baseline" period using the conflict_oil_events table. Cleaned outputs will be stored as a small set of derived CSVs, version-controlled in the project repository.

Visualization Design and Interaction

The dashboard is structured as a visual story in linked chapters, moving the user from cause to effect:

1. **The chokepoint:** a small annotated map of the Strait of Hormuz with a context panel showing the share of global oil flowing through it.
2. **The oil shock:** a time-series of Brent crude and jet fuel prices from 2019 to 2026, with conflict events overlaid as annotated markers and "Hormuz disrupted" months highlighted.
3. **The airline response:** a comparison view, using small multiples or a slope chart, showing how individual airlines change ticket prices and fuel surcharges over time, by region and route class.
4. **The passenger ticket:** a stacked "boarding-pass" breakdown showing base fare, fuel surcharge, and taxes/fees, with a before, during, and after shock comparison.
5. **The route map:** an origin-to-destination map showing rerouted, cancelled, and normal flights, with line thickness encoding extra fuel cost.

6. **Stress dashboard:** a summary view with KPI cards (oil change, average fare change, surcharges, cancellations) and filters for airline, region, route class, and conflict phase.

Interactions include time brushing, airline, region and route-class filters, hover tooltips, linked highlighting across views, and toggles between "real-world" overlays (FRED, EIA, BTS) and the "scenario" series from the Kaggle dataset.

Significance and Visual Data Science Challenges

The significance of the project lies in connecting a global geopolitical issue to an everyday passenger experience using visual storytelling, not statistical inference. The visual challenges, however, are non-trivial:

- **Mixed temporal granularities:** event-level dates, monthly oil/fuel/fare data, and quarterly financials must be aligned without misleading aggregation.
- **Multiple levels of analysis:** global oil prices, airline-level fares, route-level operations, and passenger-level receipts. The design must let users move smoothly across these scales.
- **Avoiding clutter:** multiple airlines, regions, and metrics over seven or more years require strong filtering, progressive disclosure, and careful default views.
- **Communicating a causal-looking chain without overstating causality:** the visualization must clearly show propagation patterns rather than imply deterministic cause and effect.
- **Transparency about synthetic data:** clearly distinguishing benchmark-calibrated synthetic series from real EIA, FRED and BTS series via visual cues and labelling.

Tools and Implementation

The dashboard will be implemented primarily in **Tableau Public**, which supports linked filters, coordinated views, and rapid dashboard development, and can be embedded directly in a public website. If time permits, we will use **D3.js** for one custom narrative component, most likely the scroll-driven shock-propagation timeline or the route-disruption map, to add a more distinctive storytelling element. The final website will host the embedded dashboard, an explainer video of up to 2.5 minutes, and a credits page.

Must-have Features

- Time-series of Brent crude and jet fuel with conflict event annotations
- Airline ticket-price and fuel-surcharge comparison view
- Passenger ticket breakdown (base fare, surcharge, taxes)
- At least one route-level or geographic disruption view

- Interactive filters (airline, region, route class, time period, conflict phase)
- Embedded dashboard in a public website with explainer video and credits

Optional Features

Route network map with origin-destination arcs, D3 scroll-driven storytelling component, "airline archetype" classification (cost-passers, absorbers, disrupted), aviation stress index, before-and-after shock comparison toggle, linked brushing across all views.

Schedule and Roles

Tentative plan, targeting submission on 20 May 2026.

Weeks 1 to 2 (27 April to 10 May):

Letisja Muco takes ownership of the D3 route disruption component (CH-05), the public website, and the explainer video edit, documentation.

Eirini Katrantzi takes ownership of the data preparation pipeline (date conversions, null handling, derived YoY columns) and the oil and fuel chapters (CH-01 chokepoint, CH-02 oil shock).

Alexandra Kaliakouda takes ownership of the airline response and passenger ticket chapters (CH-03 airline comparison, CH-04 ticket breakdown).

Dataset cleanup, environment setup, exploratory analysis (all members).

Weeks 3 to 4 (11 May to 17 May):

Tableau prototyping per chapter, parallelised across team members, with Saturday checkpoints to align colour scales, shared filters, and visual encodings.

Week 5 (18 May to 19 May):

Dashboard integration, the stress dashboard (CH-06) developed jointly by all three of us, website scaffolding, and explainer video recording.

Final day (20 May):

Polish, deployment, and submission.